



Hanzo Crypto Commerce: Blockchain Payment APIs and Smart Contract In- frastructure for Enterprise

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Hanzo AI

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Abstract

We describe the Hanzo crypto commerce platform, a unified API layer that enables any business to accept cryptocurrency payments, deploy smart contracts, and manage digital wallets without blockchain expertise. The platform provides zero-volatility settlement—merchants receive fiat in their bank accounts regardless of the cryptocurrency used by the customer—alongside custodial and non-custodial wallet options, audited smart contract templates for token sales and airdrops, and a management dashboard for real-time operational visibility. Built by a 2017 Techstars graduate, the system is designed on the principle that the best payment integration is one where the developer writes as little payments code as possible.

1 Introduction

By 2018, cryptocurrency adoption had reached a critical inflection point: millions of consumers held digital assets, yet the merchant infrastructure to accept those assets remained fragmented, technically demanding, and operationally risky. Price volatility alone deterred most businesses—a Bitcoin payment worth \$100 at confirmation could be worth \$85 by settlement.

Hanzo addresses this gap with a complete crypto-commerce platform whose design principle is borrowed from Stripe: *measure success by how little payments code the developer has to write*. The platform handles wallet management, transaction signing, settlement, and compliance so that engineering teams can focus on product.

2 Architecture

The platform consists of four layers:

1. **Payment Gateway.** Accepts Bitcoin, Ethereum, and ERC-20 tokens. Converts to fiat at the moment of confirmation using real-

time exchange rates, delivering zero-volatility settlement to the merchant's bank account.

2. **Wallet Services.** Provides both custodial wallets (for ease of use) and non-custodial wallets (for maximum security), selectable per user or per transaction.
3. **Smart Contract Engine.** Deploys audited, parameterized smart contracts from a template library—ICOs, airdrops, collectibles, referral programs—without requiring Solidity expertise.
4. **Management Dashboard.** A unified interface for tracking payments, managing smart contracts, monitoring token sale progress, and gaining business insights in real time.

3 Payment Processing

3.1 Zero-Volatility Settlement

The core innovation is instant fiat conversion. When a customer pays in cryptocurrency:

1. The Hanzo gateway generates a payment address and locks an exchange rate for a configurable confirmation window.
2. Upon sufficient blockchain confirmations, the gateway executes an immediate market sell on partnered exchanges.
3. The merchant receives settlement in their local currency (USD, EUR, GBP) via standard bank transfer, typically within one business day.

The merchant bears zero cryptocurrency price risk. The spread between the locked rate and the execution rate is absorbed by the platform’s liquidity reserves.

3.2 Multi-Currency Support

The payment gateway accepts any currency for which Hanzo maintains exchange liquidity. At launch: BTC, ETH, and major ERC-20 tokens. The unified API abstracts currency differences—a single integration endpoint handles all supported assets.

4 Wallet Solutions

Feature	Custodial	Non-Custodial
Key management	Hanzo HSM	User device
Recovery	Email/phone	Seed phrase
Transaction signing	Server-side	Client-side
Target user	Consumer apps	Power users

Table 1: Wallet mode comparison.

Both modes share the same API surface. Developers choose at initialization time and can offer both options to end users within the same application.

5 Smart Contract Platform

Hanzo provides audited smart contract templates covering the most common blockchain business patterns:

- **Token Sales (ICOs).** Configurable cap, pricing tiers, whitelisting, and vesting schedules.
- **Airdrops.** Batch token distribution with Merkle-proof-based claiming to minimize gas costs.
- **Collectibles.** ERC-721 non-fungible tokens with metadata management and marketplace hooks.
- **Referral Programs.** On-chain referral tracking with automatic token rewards.

Each template is parameterized via the API—no Solidity required. For custom requirements, Hanzo’s smart contract development service provides bespoke implementation with third-party audit.

6 Developer Experience

The API design follows three principles:

1. **Composable.** Small, orthogonal endpoints that combine to support any payment flow.
2. **Multi-platform.** Client libraries for web, iOS, Android, and server-side in Node.js, Python, Go, Ruby, and Haskell.
3. **Non-breaking.** Versioned API changes allow merchants to upgrade on their own schedule.

Two JavaScript libraries provide drop-in frontend integration:

- **Shop.js.** Pre-built UI components for check-out flows with validation, formatting, and browser detection.
- **Coin.js.** A lightweight library for accepting crypto payments or launching token sales with minimal frontend code.

7 Management Dashboard

The Hanzo Dashboard provides:

- Real-time charge tracking across payment types, countries, and currencies.
- Smart contract lifecycle management—deploy, monitor, pause, and upgrade.
- Token sale progress visualization with investor analytics.
- Webhook configuration for integrating payment events into existing business systems.

8 Security and Compliance

All smart contracts undergo formal verification and third-party audit before deployment. The payment gateway implements:

- Hardware security module (HSM) key storage for custodial wallets.
- Built-in fraud detection using transaction pattern analysis.
- KYC/AML integration for regulated jurisdictions.
- PCI-DSS-compliant card data handling for hybrid fiat-crypto flows.

9 Services

Beyond the self-service platform, Hanzo offers professional services:

- **White paper authoring.** Technical writing and token economics design for blockchain projects.
- **Security audit.** Smart contract and protocol review.
- **Token distribution management.** End-to-end token sale operation.

10 Conclusion

Hanzo’s crypto commerce platform eliminates the three barriers to enterprise cryptocurrency

adoption: technical difficulty, price volatility, and compliance uncertainty. By providing a Stripe-like developer experience for blockchain payments, audited smart contract templates, and zero-volatility settlement, the platform enables any business to participate in the cryptocurrency economy without specialized blockchain expertise. Hanzo is dedicated to advancing decentralization by making crypto payments accessible, opening the door for the world to transact on the blockchain.

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